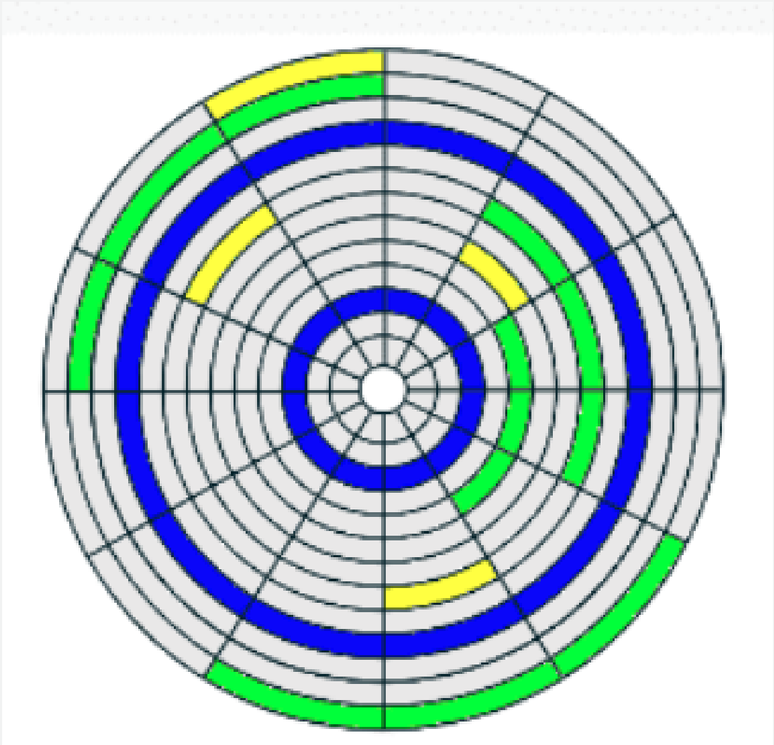
GESTOR: MEGATRON3000

Presentado por: Coloma Yujra Riki Santher

ESTRUCTURA DE DISCO

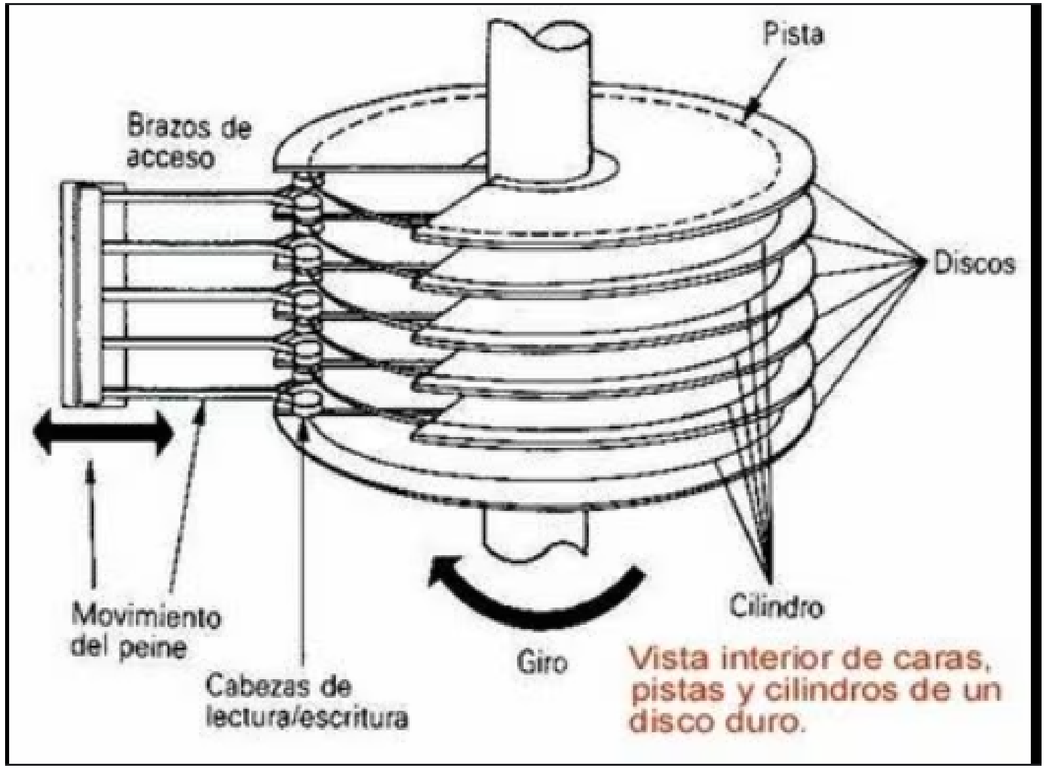


```
1 Platos: 2
2 Caras por plato: 2
3 Pistas por cara: 10
4 Sectores por pista: 16
5 Tamaño sector: 512
```

```
disco  include  obj  src  DISCO  Makefile
❯ cd disco
❯ ls
Plato0  Plato1  header.txt
❯ tree
.
|-- Plato0
|   |-- Cara0
|   |   |-- Pista0
|   |   |   |-- Sector0
|   |   |   |-- Sector1
|   |   |   |-- Sector10
|   |   |   |-- Sector11
|   |   |   |-- Sector12
|   |   |   |-- Sector13
|   |   |   |-- Sector14
|   |   |   |-- Sector15
|   |   |   |-- Sector2
|   |   |   |-- Sector3
|   |   |   |-- Sector4
|   |   |   |-- Sector5
|   |   |   |-- Sector6
|   |   |   |-- Sector7
|   |   |   |-- Sector8
|   |   |   `-- Sector9
|   |   |-- Pista1
|   |   |   |-- Sector0
|   |   |   |-- Sector1
|   |   |   |-- Sector10
|   |   |   |-- Sector11
|   |   |   |-- Sector12
|   |   |   |-- Sector13
|   |   |   |-- Sector14
|   |   |   |-- Sector15
|   |   |   |-- Sector2
|   |   |   |-- Sector3
|   |   |   |-- Sector4
|   |   |   |-- Sector5
```

```
Plato0  Plato1  header.txt
❯ cd Plato0/Cara0/Pista0
❯ ls
Sector0  Sector10  Sector12  Sector14  Sector2  Sector4  Sector6  Sector8
Sector1  Sector11  Sector13  Sector15  Sector3  Sector5  Sector7  Sector9
```

Capacidad del disco



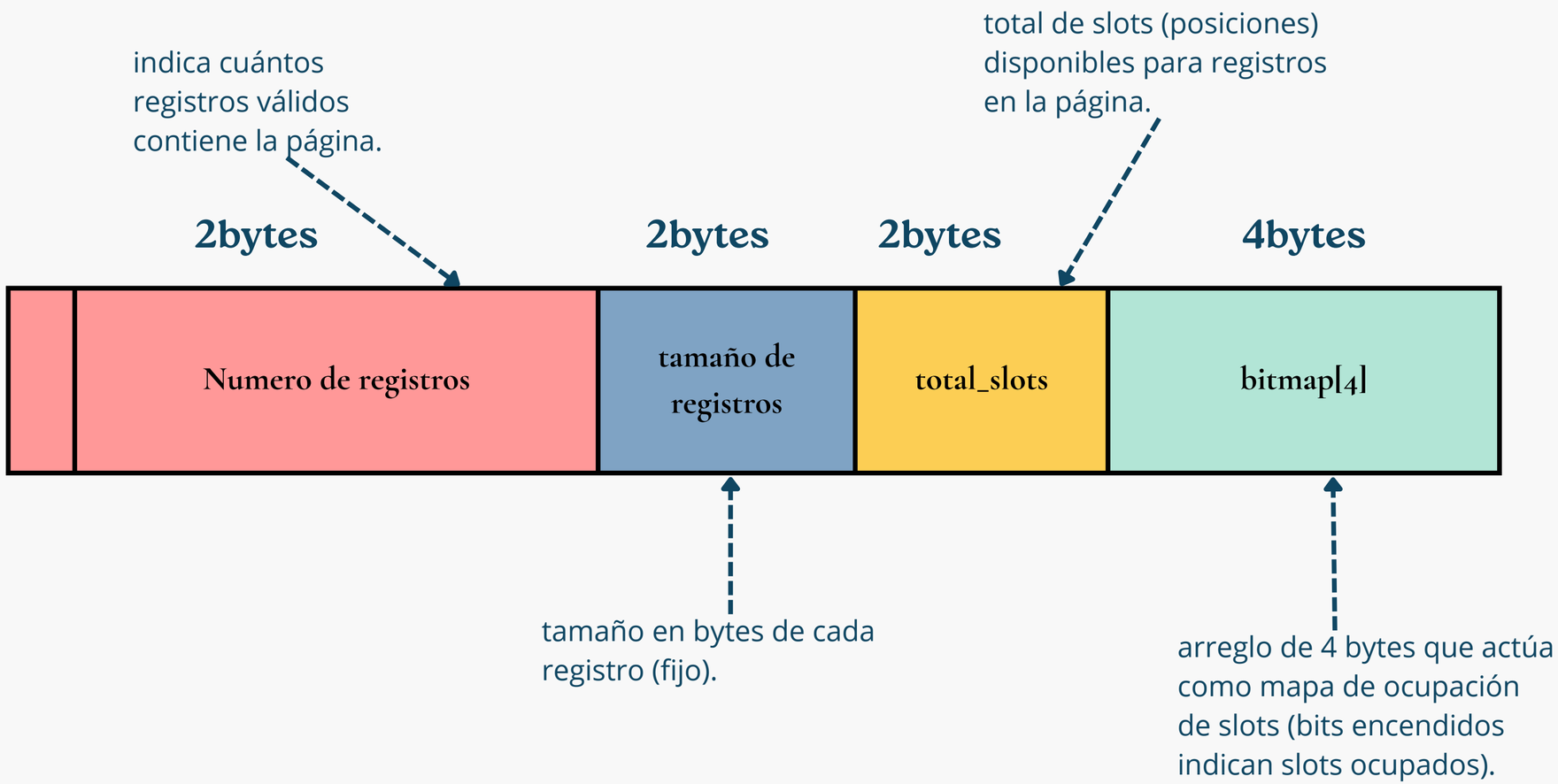

```
-----------------------------------------------
Opción: 4
Cantidad actual de platos: 2
Pistas por cara: 10
Sectores por pista: 16
Tamaño total del disco: 327680 bytes

DISCO VIRTUAL
-----------------------------------------------
1. Agregar platos
2. Modificar pistas y sectores en todos los platos
3. Eliminar plato
4. Mostrar capacidad del disco
0. Salir
-----------------------------------------------
Opción:
```

```
1 titanic6#PassengerId#int#Survived#int#Pclass#int#Name#string#Sex#string#Age#int#SibSp#int#Parch#int#Ticket#string#Fare#float#Ca
  bin#string#Embarked^M#string#variable
```

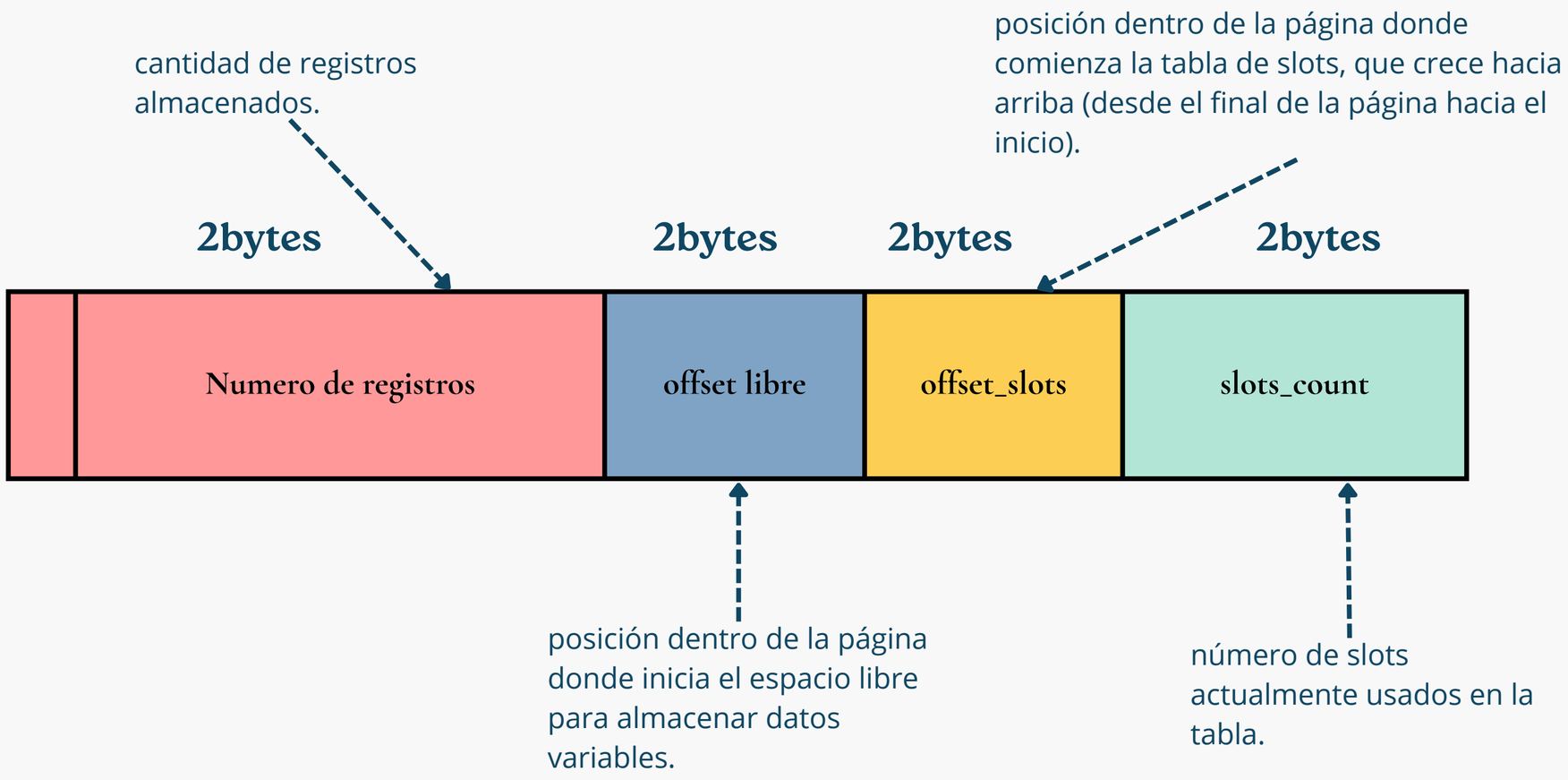
Registros de Longitud Fija y Variable

```cpp
#ifndef DISK_CONSTANTS_H
#define DISK_CONSTANTS_H

#include <cstdint>

#define MAX_PISTAS 10
#define MAX_PLATOS 2
#define MAX_CARAS 2
#define MAX_SECTORES 16


#define SECTOR_SIZE 512
#define HOUSING_RECORD_SIZE 64
#define TITANIC_RECORD_SIZE 128
#define PAGE_SIZE 2048
#define SECTORS_PER_PAGE 4

struct CabeceraFija {
    uint16_t num_registros;
    uint16_t tam_registro;
    uint16_t total_slots;
    uint8_t bitmap[4];
};

struct CabeceraVariable {
    uint16_t num_registros;
    uint16_t offset_libre;      // Donde empieza el espacio libre para datos
    uint16_t offset_slots;      // Donde empieza la tabla de slots (crece hacia arriba)
    uint16_t slots_count;       // Número actual de slots
    // Los slots se almacenan en la página, no aquí fijo
};




#endif
```

Estructura de la cabecera para registros de longitud fija



Estructura de la cabecera para registros de longitud variable



ESQUEMA

```
1 titanicG#PassengerId#int#Survived#int#Pclass#int#Name#string#Sex#strin
  g#Age#int#SibSp#int#Parch#int#Ticket#string#Fare#float#Cabin#string#Em
  barked^M#string#variable
2 Housing#price#int#area#int#bedrooms#int#bathrooms#int#stories#int#main
  road#string#guestroom#string#basement#string#hotwaterheating#string#ai
  rconditioning#string#parking#int#prefarea#string#furnishingstatus^M#st
  ring#fijo
```

Página con registros de tamaño variable

Alacritty

```
1 ^\^@<9e>^GÈ^G^\^@1#0#3#Braund, Mr. Owen Harris#male#22#1#0#A/5 21171#7.25##S
2 2#1#1#Cumings, Mrs. John Bradley (Florence Briggs Thayer)#female#38#1#0#PC 17599#71.2833#C85#C
3 3#1#3#Heikkinen, Miss. Laina#female#26#0#0#STON/O2. 3101282#7.925##S
4 4#1#1#Futrelle, Mrs. Jacques Heath (Lily May Peel)#female#35#1#0 #113803#53.1#C123#S
5 5#0#3#Allen, Mr. William Henry#male#35#0#0 #373450#8.05##S
6 6#0#3#Moran, Mr. James#male##0#0 #330877#8.4583##Q
7 7#0#1#McCarthy, Mr. Timothy J#male#54#0#0#17463#51.8625#E46#S
8 8#0#3#Palsson, Mast
```

Alacritty

```
er. Gosta Leonard#male#2#3#1#349909#21.075##S
9#1#3#Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)#female#27#0#2#347742#11.1333##S
10#1#2#Nasser, Mrs. Nicholas (Adele Achem)#female#14#1#0#237736#30.0708##C
11#1#3#Sandstrom, Miss. Marguerite Rut#female#4#1#1#PP 9549#16.7#G6#S
12#1#1#Bonnell, Miss. Elizabeth#female#58#0#0#113783#26.55#C103#S
13#0#3#Saundercock, Mr. William Henry#male#20#0#0#A/5. 2151#8.05##S
14#0#3#Andersson, Mr. Anders Johan#male#39#1#5#347082#31.275##S
15#0#3#Vestrom, Miss. Hulda
```

Información de la cabecera

```
❯ hexdump -C Sector0
00000000  1c 00 9e 07 c8 07 1c 00  31 23 30 23 33 23 42 72  |........1#0#3#Br|
00000010  61 75 6e 64 2c 20 4d 72  2e 20 4f 77 65 6e 20 48  |aund, Mr. Owen H|
00000020  61 72 72 69 73 23 6d 61  6c 65 23 32 32 23 31 23  |arris#male#22#1#|
00000030  30 23 41 2f 35 20 32 31  31 37 31 23 37 2e 32 35  |0#A/5 21171#7.25|
00000040  23 23 53 0d 0a 32 23 31  23 31 23 43 75 6d 69 6e  |##S..2#1#1#Cumin|
00000050  67 73 2c 20 4d 72 73 2e  20 4a 6f 68 6e 20 42 72  |gs, Mrs. John Br|
00000060  61 64 6c 65 79 20 28 46  6c 6f 72 65 6e 63 65 20  |adley (Florence |
00000070  42 72 69 67 67 73 20 54  68 61 79 65 72 29 23 66  |Briggs Thayer)#f|
00000080  65 6d 61 6c 65 23 33 38  23 31 23 30 23 50 43 20  |emale#38#1#0#PC |
00000090  31 37 35 39 39 23 37 31  2e 32 38 33 33 23 43 38  |17599#71.2833#C8|
000000a0  35 23 43 0d 0a 33 23 31  23 33 23 48 65 69 6b 6b  |5#C..3#1#3#Heikk|
000000b0  69 6e 65 6e 2c 20 4d 69  73 73 2e 20 4c 61 69 6e  |inen, Miss. Lain|
000000c0  61 23 66 65 6d 61 6c 65  23 32 36 23 30 23 30 23  |a#female#26#0#0#|
000000d0  53 54 4f 4e 2f 4f 32 2e  20 33 31 30 31 32 38 32  |STON/O2. 3101282|
000000e0  23 37 2e 39 32 35 23 23  53 0d 0a 34 23 31 23 31  |#7.925##S..4#1#1|
000000f0  23 46 75 74 72 65 6c 6c  65 2c 20 4d 72 73 2e 20  |#Futrelle, Mrs. |
```

Final de página

```
ss. Anna ""Annie""#female#15#0#0#330923#8.0292##Q
24#1#1#Sloper, Mr. William Thompson#male#28#0#0#113788#35.5#A6#S
25#0#3#Palsson, Miss. Torborg Danira#female#8#3#1#349909#21.075##S
26#1#3#Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johansson)#female#38#1#5#347077#31.3875##S
27#0#3#Emir, Mr. Farred Chehab#male##0#0#2631#7.225##C
28#0#1#Fortune, Mr. Charles Alexander#male#19#3#2#19950#263#C23 C25 C27#S
                                        S^G^[^G¹^Fu^F3^Fî^E´^E~^ED^Eé^D-^Dt^D0^Dä^C£^C^^C^[^CÔ^
B<88>^B/^Bí^A®^A{^A@^Aë^@¥^@E^@^H^@
```

Páginas con registro de tamaño Fijo para Housing

Alacritty

```
1 ^_^@@^@^_^@ÿÿÿ^?13300000#7420#4#2#3#yes#no#no#no#yes#2#yes#furnished^M
2 12250000#8960#4#4#4#yes#no#no#no#yes#3#no#furnished^M
3 12250000#9960#3#2#2#yes#no#yes#no#no#2#yes#semi-furnished^M
4 12215000#7500#4#2#2#yes#no#yes#no#yes#3#yes#furnished^M
5 11410000#7420#4#1#2#yes#yes#yes#no#yes#2#no#furnished^M
6 10850000#7500#3#3#1#yes#no#yes#no#yes#2#yes#semi-furnished^M
7 10150000#8580#4#3#4#yes#no#no#no#yes#2#yes#semi-furnished^M
8 10150000#16200#5#3#2#yes#no#no#no#no#0#no#unfurnished^M
```

Otro ejemplo

```
1 ^_^@@^@^_^@ÿÿÿ^?2695000#4000#2#1#1#yes#no#no#no#no#0#no#unfurnished^M
2 2660000#3934#2#1#1#yes#no#no#no#no#0#no#unfurnished^M
3 2660000#2000#2#1#2#yes#no#no#no#no#0#no#semi-furnished^M
4 2660000#3630#3#3#2#no#yes#no#no#no#0#no#unfurnished^M
5 2660000#2800#3#1#1#yes#no#no#no#no#0#no#unfurnished^M
6 2660000#2430#3#1#1#no#no#no#no#no#0#no#unfurnished^M
7 2660000#3480#2#1#1#yes#no#no#no#no#1#no#semi-furnished^M
8 2660000#4000#3#1#1#yes#no#no#no#no#0#no#semi-furnished
```

Información de la cabecera de registros de tamaño fijo

```
Sector0   Sector10   Sector12   Sector14   Sector2   Sector4   Sector6   Sector8
Sector1   Sector11   Sector13   Sector15   Sector3   Sector5   Sector7   Sector9
❯ hexdump -C Sector0
00000000  1f 00 40 00 1f 00 ff ff  ff 7f 32 36 39 35 30 30  |..@.......269500|
00000010  30 23 34 30 30 30 23 32  23 31 23 31 23 79 65 73  |0#4000#2#1#1#yes|
00000020  23 6e 6f 23 6e 6f 23 6e  6f 23 6e 6f 23 30 23 6e  |#no#no#no#no#0#n|
00000030  6f 23 75 6e 66 75 72 6e  69 73 68 65 64 0d 20 20  |o#unfurnished.  |
00000040  20 20 20 20 20 20 20 20  20 0a 32 36 36 30 30 30  |         .266000|
00000050  30 23 33 39 33 34 23 32  23 31 23 31 23 79 65 73  |0#3934#2#1#1#yes|
00000060  23 6e 6f 23 6e 6f 23 6e  6f 23 6e 6f 23 30 23 6e  |#no#no#no#no#0#n|
00000070  6f 23 75 6e 66 75 72 6e  69 73 68 65 64 0d 20 20  |o#unfurnished.  |
00000080  20 20 20 20 20 20 20 20  20 0a 32 36 36 30 30 30  |         .266000|
00000090  30 23 32 30 30 30 23 32  23 31 23 32 23 79 65 73  |0#2000#2#1#2#yes|
000000a0  23 6e 6f 23 6e 6f 23 6e  6f 23 6e 6f 23 30 23 6e  |#no#no#no#no#0#n|
000000b0  6f 23 73 65 6d 69 2d 66  75 72 6e 69 73 68 65 64  |o#semi-furnished|
```

Lectura de varias páginas

Alacritty

price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus	fijo
13300000	7420	4	2	3	yes	no	no	no	yes	2			
12250000	8960	4	4	4	yes	no	no	no	yes	3			
12250000	9960	3	2	2	yes	no	yes	no	no	2		furnished	
12215000	7500	4	2	2	yes	no	yes	no	yes	3			
11410000	7420	4	1	2	yes	yes	yes	no	yes	2			
10850000	7500	3	3	1	yes	no	yes	no	yes	2		furnished	
10150000	8580	4	3	4	yes	no	no	no	yes	2		furnished	
10150000	16200	5	3	2	yes	no	no	no	no	0		shed	
9870000	8100	4	1	2	yes	yes	yes	no	yes	2			
9800000	5750	3	2	4	yes	yes	no	no	yes	1		shed	
9800000	13200	3	1	2	yes	no	yes	no	yes	2			
9681000	6000	4	3	2	yes	yes	yes	yes	no	2		furnished	
9310000	6550	4	2	2	yes	no	no	no	yes	1		furnished	
9240000	3500	4	2	2	yes	no	no	yes	no	2			
9240000	7800	3	2	2	yes	no	no	no	no	0		furnished	
9100000	6000	4	1	2	yes	no	yes	no	no	2		furnished	
9100000	6600	4	2	2	yes	yes	yes	no	yes	1		shed	
8960000	8500	3	2	4	yes	no	no	no	yes	2			
8890000	4600	3	2	2	yes	yes	no	no	yes	2			
8855000	6420	3	2	2	yes	no	no	no	yes	1		furnished	
8750000	4320	3	1	2	yes	no	yes	yes	no	2		furnished	
8680000	7155	3	2	1	yes	yes	yes	no	yes	2		shed	
8645000	8050	3	1	1	yes	yes	yes	no	yes	1			
8645000	4560	3	2	2	yes	yes	yes	no	yes	1			
8575000	8800	3	2	2	yes	no	no	no	yes	2			
8540000	6540	4	2	2	yes	yes	yes	no	yes	2			
8463000	6000	3	2	4	yes	yes	yes	no	yes	0		furnished	
8400000	8875	3	1	1	yes	no	no	no	no	1		furnished	
8400000	7950	5	2	2	yes	no	yes	yes	no	2		shed	
8400000	5500	4	2	2	yes	no	yes	no	yes	1		furnished	
8400000	7475	3	2	4	yes	no	no	no	yes	2		shed	
8400000	7000	3	1	4	yes	no	no	no	yes	2		furnished	

cpp

E.md

E.md

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Archivos clave para el funcionamiento

```
❯ ls -R
build   docs    schema   tests    Makefile    README.md
data    include src      LICENSE  megatron3000

./build:
conditionalSelect.o  main.o     query.o
loader.o             printer.o  simpleSelect.o

./data:
datasecTitanic.tsv  Housing.txt   titanicG.txt
Housing.csv         titanicG.csv

./docs:

./include:
disk_constants.h  loader.h  printer.h  query.h

./schema:
schema.txt

./src:
conditionalSelect.cpp  main.cpp     query.cpp
loader.cpp             printer.cpp  simpleSelect.cpp
```